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NODIRBEK MAKHTUMOV

AI/ML Engineer

Computer Vision · Edge AI · Model Optimization

E-7 Special Ability · Authorized to work in Korea

Summary

AI/ML engineer with 3+ years building production computer vision systems that run on edge hardware. I own the full path: model design and training, quantization and TensorRT/ONNX conversion, deployment to NVIDIA Jetson, Hailo-8 and Android devices, FastAPI services and Docker-based edge operation. I validate models with per-class metrics, adversarial robustness testing and per-hardware measurement, not a single headline accuracy figure. I also build LLM-based agents with human approval and evaluation.

Selected Results

30%	25+ FPS	8.6×	76% → 92%
GPU memory reduction	4 concurrent camera streams	NPU inference speedup	Truck-ID OCR accuracy

Technical Stack

Computer Vision	Object Detection, Semantic / Instance Segmentation, OCR, Multi-Object Tracking, Face Recognition (ArcFace), Pose Estimation
LLM & Agents	Claude API (structured outputs, prompt caching, Batch API), LLM evaluation, cross-encoder re-ranking
Frameworks	PyTorch, TensorFlow, Keras, Scikit-Learn, Ultralytics (YOLO), OpenCV, InsightFace, Hugging Face Transformers, SAM2, sherpa-onnx (speech-to-text)
Optimization	Quantization, Pruning, TensorRT, ONNX Runtime, DLA offloading, backbone replacement and A/B evaluation
Edge Hardware	NVIDIA Jetson Orin NX / Orin Nano, Hailo-8 NPU, RK3588 NPU (Orange Pi 5), Raspberry Pi 4/5, Android tablets
Deployment	Docker, FastAPI, Flask, REST API, PostgreSQL, SQLite, concurrent multi-stream serving
Evaluation	Per-class metrics, adversarial robustness testing, Grad-CAM, FPS / latency / jitter benchmarking, W&B, MLflow, TensorBoard
Languages & Tools	Python (primary), C++, SQL, Linux (Ubuntu), Git/GitHub, GitLab, GitHub Actions CI

Experience

Netvision Telecom — AI/ML Engineer · Software Development Team — 2023.04 – Present

Daejeon · Joined as a paid research intern in April 2023, converted to full-time in September 2023

- **Smart waste collection fleet management** — YOLO waste classification and load-volume estimation on the vehicle. Replaced the YOLOv8 CSPDarknet backbone with ResNet50, 1x1 conv adapters (512/1024/2048 → 192/384/576) into the PANet neck. ResNet layers run on the Jetson DLA: **~30% less GPU memory on Jetson Orin NX**, so vision and fleet management share one board. A/B tested mAP, FPS, memory on-device; quantization, TensorRT, Docker
- **Edge-aware segmentation pipeline (rail-wheel gap measurement)** — Lightweight segmentation model, RANSAC line fitting for the rail edge and a custom temporal point-stabilization algorithm for the curved wheel edge. Multi-threaded, one queue per stream; FastAPI, PostgreSQL logging: **25+ FPS across 4 concurrent camera streams**
- **SAM2 automated video labeling tool** — From one first-frame prompt (box or mask), SAM2 labels the whole video; a post-check looks for occlusion, drift and target loss; export to YOLO and COCO. **10x+ faster dataset construction**
- **(Research intern, 2023.04–2023.09)** Built and deployed YOLO models on Jetson Orin Nano: latency optimization, quantization, dataset collection, normalization and labeling

Experience (continued)

Freelance contract AI/ML Engineer — Email quoting agent (Telegram approval) 2026.10
github.com/Nodir0705/sales-quote-agent

- Drafts replies to price enquiries for a company that exports insulated sandwich panels. An LLM reads each email into a fixed schema; **Python computes the quote** from the Excel price list (discount tiers, containers, freight), so the model never types a price. An invented figure blocks the draft; nothing is sent until the owner taps **Approve on Telegram**
- Claude API (structured outputs, prompt caching, Batch API), SQLite. On written test emails: **extraction eval 26/26**, **1,513 automated checks** passing. Built and run against a real inbox and Telegram in test mode

Freelance contract AI/ML Engineer — Face recognition attendance kiosk 2026.07
github.com/Nodir0705/face-recognition

- Same pipeline built three ways (**Python/CPU, C++ port, Hailo-8 NPU**) to choose hardware by measurement. Raspberry Pi 5, detect + embed per face, CPU vs Hailo-8: median 48.08 ms → 5.59 ms (**8.6× faster**), **95× lower jitter**
- C++ port, on a Raspberry Pi 4, only cut loop overhead (~10 ms → ~1 ms): the model dominates, so the NPU is the lever
- SCRFD + ArcFace 512-d, cosine match over 3 consecutive frames, blink + motion liveness. Delivered on-device on a **Samsung Galaxy Tab** (Android, Kotlin, TFLite); offline-first SQLite with Google Sheets sync

Freelance contract AI/ML Engineer — Smart truck parking & truck-ID recognition 2025.09 – 2025.11
github.com/Nodir0705/smart-truck-parking

- Built **end-to-end, solo**: YOLO finds trucks and ID plates, ByteTrack tracks them, line crossing gives IN / OUT events, IDs voted over the last 5 reads. A **deskew stage** before EasyOCR targeted the main failure, perspective distortion: **ID accuracy 76% → 92%** on the client's cameras
- Open-source version with a rewritten deskew module: on 200 synthetic tilted plates, EasyOCR accuracy **80.5% → 97.5%**

Projects & Other Activities

Smart Pointer — speech-driven slide highlighting · Personal project · 2026 · github.com/Nodir0705/smart-pointer
Highlights the part of the slide a presenter is talking about. The live loop runs offline on a CPU: speech-to-text → clauses → a **2-layer TinyBERT** re-ranker (**16 ms** per decision, one CPU thread) → highlight. **77.3% of pointing steps lit** on 7 unseen public decks, scored with a synthetic TTS voice; not yet measured on a real presentation

IoT security gateway (Orange Pi 5 · RK3588S) · Personal project · 2026 · github.com/Nodir0705/iot-gateway-security
No cloud; 4 detection layers: XDP/eBPF, **244 custom Suricata IoT rules**, NPU flow autoencoder, behavior rules. 47-minute test, 9 attack phases: **9/9 phases detected**, **87/88 (98.9%)** expected rules fired. Lab use only (NSL-KDD, synthetic flows)

Adversarial robustness of a malware image classifier · Personal project · 2026 · github.com/Nodir0705/malware_classifier
ResNet50 on MalImg byteplots (25 families): 99.48% clean accuracy falls to 22% under PGD-20 (each byte changed by at most 2). Full-network adversarial retraining restored **94.15% under attack** for a 2.3-point clean-accuracy cost

Mentor · Woosong University Capstone · 2024
Mentored a computer vision team (image processing, YOLO training) with weekly feedback; the team placed 1st of 200+ teams

Education

BSc, Computer Science Woosong University · Daejeon 2019 – 2023

Languages

Uzbek	Native	**English**	Fluent — professional working proficiency
Korean	Conversational	**Russian**	Conversational

Visa

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